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Resume Enhancing, Analyzing and Developing using Machine Learning (R.E.A.D.M.L.)

Mushan R. A. [*1], Sankanna A. S.[*2], Chidgumpi A. S.[*3], Surwase V. V.[*4], Prof Pasnur P. S.[*5]

[*1,2,3,4,5] Dept. Of Artificial Intelligence & Data Science, V. V. P. Institute of Engineering and Technology, Solapur, Maharashtra, India.

ABSTRACT

R.E.A.D.M.L. is an advanced AI-driven platform designed to enhance the process of resume development and optimization. Built using Python, Streamlit, Natural Language Processing (NLP), and the Gemini API, the system empowers users to either create a new resume from scratch or analyse an existing one. Unlike conventional resume-building tools that focus primarily on layout, R.E.A.D.M.L. offers a comprehensive solution by enabling real-time resume generation, ATS (Applicant Tracking System) scoring, and personalized feedback tailored to each user's profile. With its robust architecture, intelligent document processing, and interactive data visualizations, R.E.A.D.M.L. helps users gain clear insights into their resume's effectiveness, making it highly relevant in today's digital recruitment landscape.

Keywords: Resume Analysis, Python, Streamlit, Machine Learning, Natural Language Processing, ATS Score, Gemini API, AI feedback, Resume developing, Data Visualization.

I. INTRODUCTION

In the era of rising demand for AI and Machine Learning in recruitment, there is a growing need to automate tasks that reduce both time and effort. Traditional resume-building websites primarily focus on layout and design, which often creates problems for candidates when their resumes are evaluated by Applicant Tracking Systems (ATS). R.E.A.D.M.L. addresses this challenge by integrating intelligent algorithms that not only assist in developing a resume but also in analysing it for ATS compatibility. Leveraging modern AI and Natural Language Processing (NLP) techniques, R.E.A.D.M.L. provides real-time feedback to help users identify gaps and improve their resumes effectively. The development of R.E.A.D.M.L. – Resume Enhancing, Analysing & Development using Machine Learning marks a significant advancement in the way resumes are created and evaluated, combining the power of AI and NLP to deliver smarter, more impactful results. A significant step forward in creating and analyzing resumes by integrating the advanced AI & NLP techniques which plays key role in the R.E.A.D.M.L.

II. METHODOLOGY

The Development and execution of R.E.A.D.M.L. followed a systematic and structured multiphase methodology which integrates principles from the software engineering, machine learning and also the user experience design

Technical Stack:

- Frontend: Streamlit for UI development
- Backend: Python, custom NLP modules, Gemini API integration
- Document/PDF Processing: wkhtmltopdf, python-docx, PyPDF2
- Visualization: Plotly for good and interactive charts

III. MODELING AND ANALYSIS

The R.E.A.D.M.L. provides architecture separates the data gathered, parsed and AI analysis on the parsed data and generates the resume document this process ensures the maintainability and scalability. In the resume parser it utilizes the NLP to identify the sections and extract relevant data for the section. The sections are parsed to the Gemini API which is used as an intelligent resume reviewer which proves the accurate analysis and suggestion of the resume parsed.

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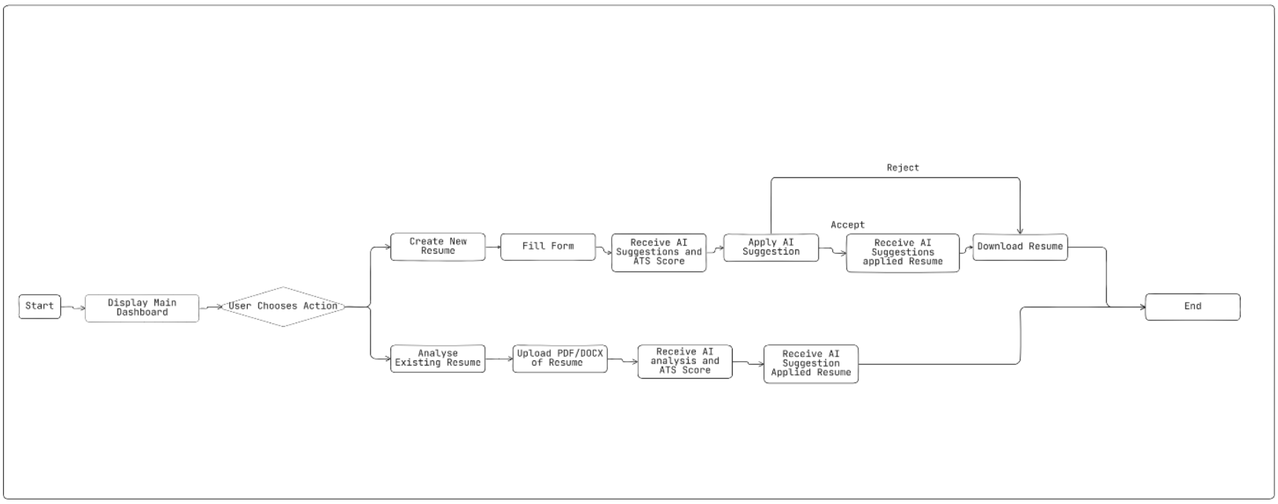

Figure 1: workflow/activity diagram

The Visualization is used to convert the ATS score and other analysis section to the radar, pie chart making user for better experience.

IV. RESULTS AND DISCUSSION

Testing the R.E.A.D.M.L. which effective advises the key changes, missing skills and format issues that need to be solved/fixed. The ATS system correctly matches the industry benchmarks, the design allows user for easy-to-use advantage without any prior knowledge. The future implementations are such as resume building using the chatbot, from linkedin link to users resume building and many other implementations. It also auto analyzes the resume after the creating the resume from our R.E.A.D.M.L. websites which allows user to know in which section the gap is to fill it the analysis is required.

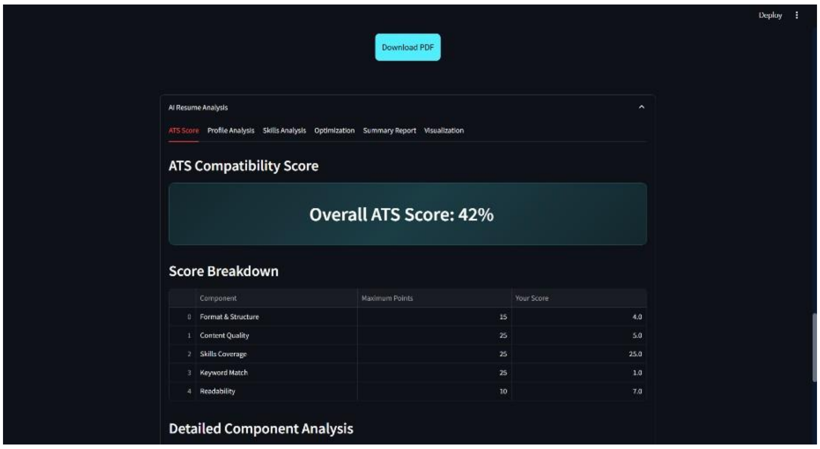

Figure 2: Output of Analysis

V. CONCLUSION

This paper presents the successful development of **R.E.A.D.M.L.**, a Python-based platform built with Streamlit, spaCy, NLP, and machine learning techniques. The system introduces a significant improvement in the domain of resume creation and analysis by offering AI-powered feedback through an intuitive and user-friendly interface. Unlike traditional tools, R.E.A.D.M.L. not only generates well-structured, ATS-compliant resumes but also delivers detailed insights to help users identify and address gaps in their profiles—thereby increasing their chances of passing ATS filters. Looking ahead, future enhancements will focus on expanding the AI capabilities of the platform and further improving the accuracy and personalization of its analysis.

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